

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 22-Mar-2021

Time: Three Hours

Max. Marks: 75 Marks

PHYSICAL PHARMACEUTICS - II

Q.P. CODE: 5015

Your answers should be specific to the questions asked.

Draw neat labeled diagrams wherever necessary.

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. List out various derived properties of powder and explain the method to determine specific surface area of powder by gas displacement method.

OR

Deduce an equation for first order reaction rate constant. Show the inter relationship between the initial concentration and the half period in first order reaction.

2. Define Thixotrophy. Explain the different methods of measurement of Thixotrophy.

SHORT ESSAYS

7 x 5 = 35 Marks

3. Write any two methods of purification of colloids.

OR

Describe association colloids. Mention their application.

4. Define plug flow and how it can be overcome. Mention the advantages of a cone and plate viscometer.

OR

Explain dilatant systems with suitable example.

5. Define deformation. Explain elastic and plastic deformation of solids.
6. Discuss the factors affecting settlement of particle in suspensions.
7. Explain any two physical instability parameters in an emulsion.
8. Define angle of repose and discuss the method to determine it. Give its significance.
9. How do you protect drugs against oxidative breakdown?

SHORT ANSWERS

10 x 2 = 20 Marks

10. Define Hardy-Schulze rule.
11. Why colloidal solutions are coloured?
12. Define photolysis and zero order reaction
13. Give the importance of bulges and spurs.
14. Define poise.
15. Define degree of flocculation.
16. Write the applications of bulk density and give its equation.
17. Define porosity. Give its applications in pharmacy.
18. Define half-life and shelf life of a reaction.
19. Define energy of activation and give its importance.
